

Tailoring Machine Learning Weather Predictions for Local Impacts

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Research Goal and Overview

- Show the feasibility of finetuning (localizing) weather forecasts for specific outcomes using minimal compute power
- Demonstrate the method using 2 meter temperature and 10 meter wind speed across a variety of regions and lead times
- Show that an ML forecast like Pangu can be corrected in the same way as traditional NWP forecast like IFS

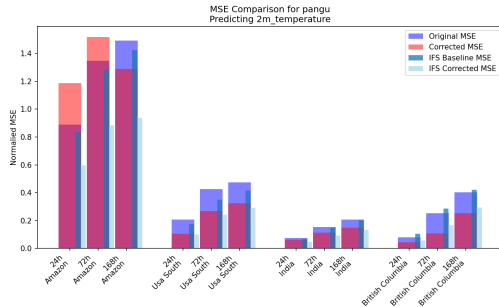
Key Contribution: Demonstrating a simple method to significantly improve forecast accuracy using limited computational power.

Method

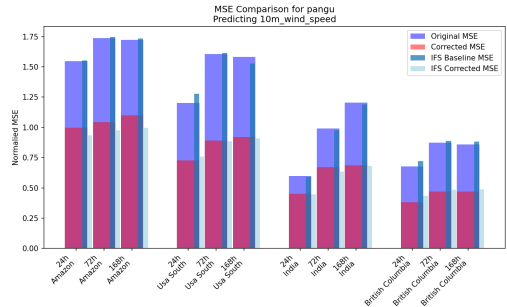
- Uses Pangu and IFS HRES forecasts with lead times from 1 to 7 days
 - Target Pangu predictions using ERA5 and IFS HRES using its $t=0$ initialization
- Filter to a single variable and lead time, in a particular region
- Train simple MLP network that takes in Pangu/IFS forecast outputs and minimizes the forecast error
 - Train using a single forecast per day for years 2018-2021
 - Test using daily forecasts from 2022

Headline Result: Simple MLP Seems Effective at Correcting Forecasts

- Simple MLP method works across variety of regions and climates
- Training a correction model for a 10x10 degree region takes about 2.5 minutes on an M3 max MacBook Pro
- Similar improvements to both Pangu and IFS forecasts

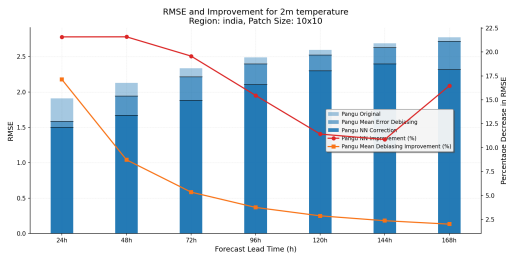


(a) 2m Temperature

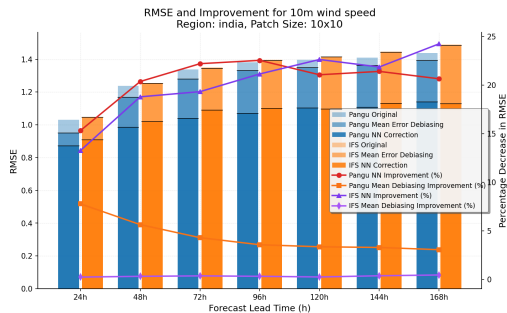


(b) Wind Speed

See Similar Percent Improvements Across Different Lead Times



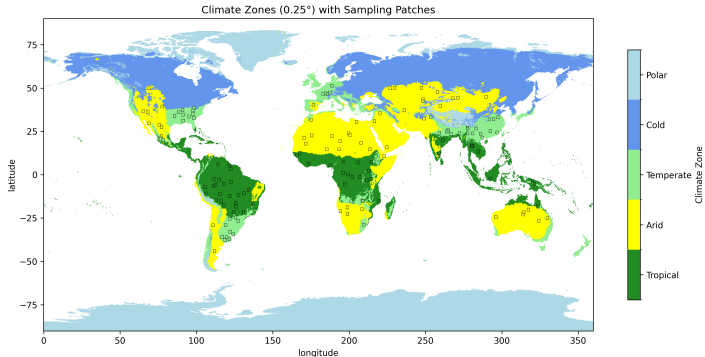
(a) 2m Temperature: 10x10 degree



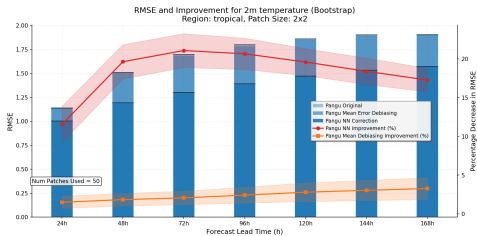
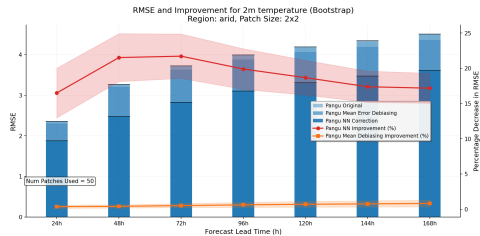
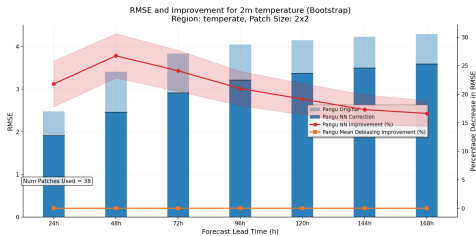
(b) Wind Speed: 10x10 degree

How to Explore Heterogeneity by Climate Region

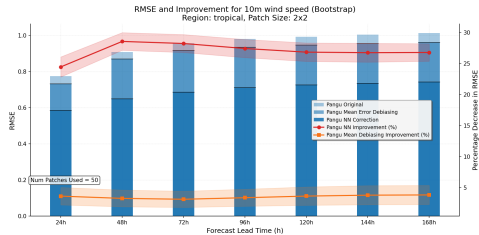
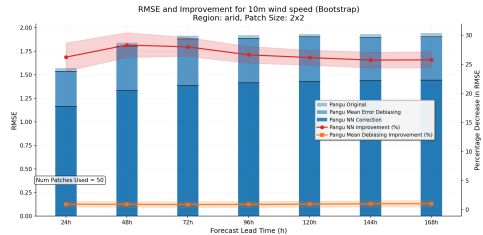
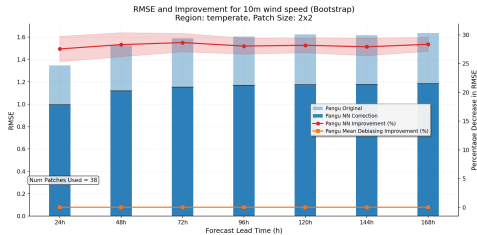
- Define tropical, temperate, and arid pixels on an 0.25 degree grid using broad Koppen-Geiger classes
- Train fine-tuning model on randomly sampled 2x2 degree patches



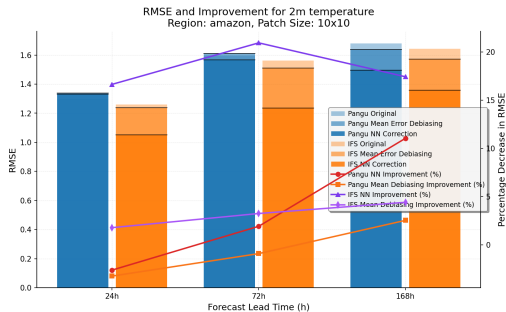
Heterogeneity by Climate Region: 2m Temperature



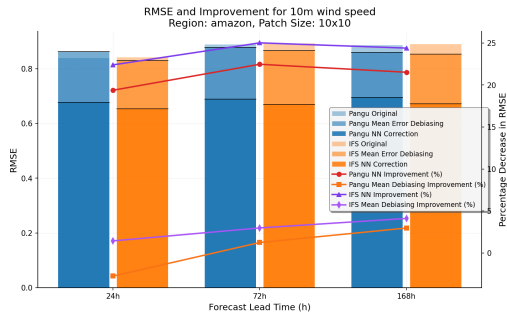
Heterogeneity by Climate Region: 10m Wind Speed



Amazon

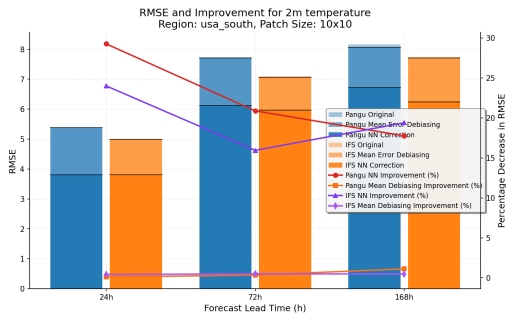


(a) 2m Temperature: 10x10 degree

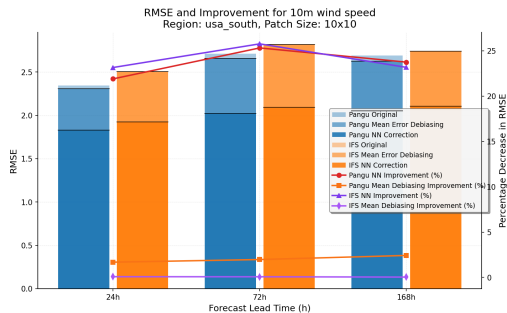


(b) Wind Speed: 10x10 degree

US South

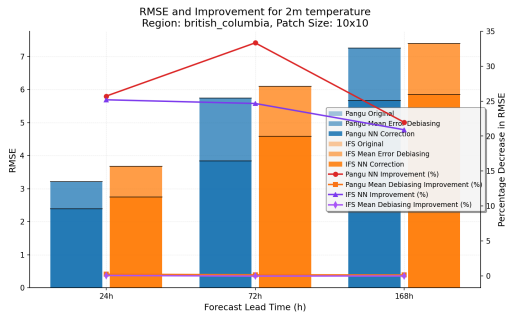


(a) 2m Temperature: 10x10 degree

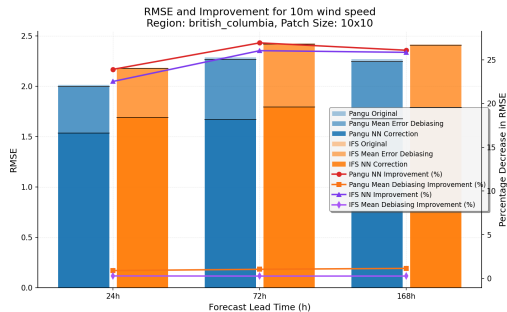


(b) Wind Speed: 10x10 degree

British Columbia



(a) 2m Temperature: 10x10 degree



(b) Wind Speed: 10x10 degree

Previous Paper Corrected IFS Used a U-net

- [?] trained a U-Net to correct IFS Forecasts in Northern China using 12 years of data and a NVIDIA GPU
 - Outperforms traditional method of correcting by subtracting mean forecast error
- Uses RMSE error instead of MSE but I find a larger percent improvement for both 2m temperature and 10m wind speed

What I'm Actively Working on

- Rewriting the databuild to be able to take advantage of the new WeatherbenchX package and also be able to run on a server and be more flexible
- Implementing a U-Net model for fine-tuning
 - Early results suggest it significantly ups compute time without much improved performance
- Implement ANO Correction (subtracting mean error) method to compare to MLP correction